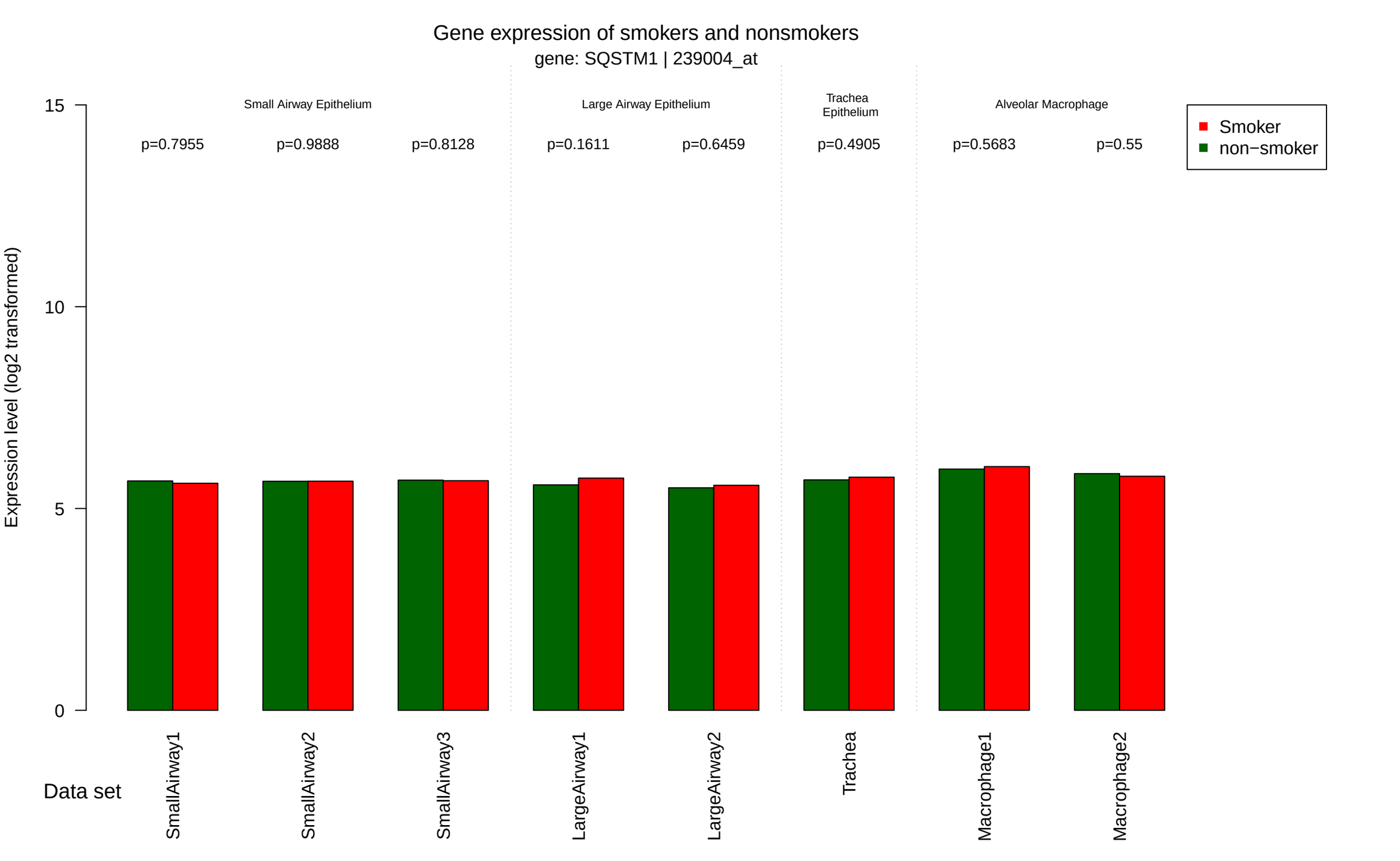
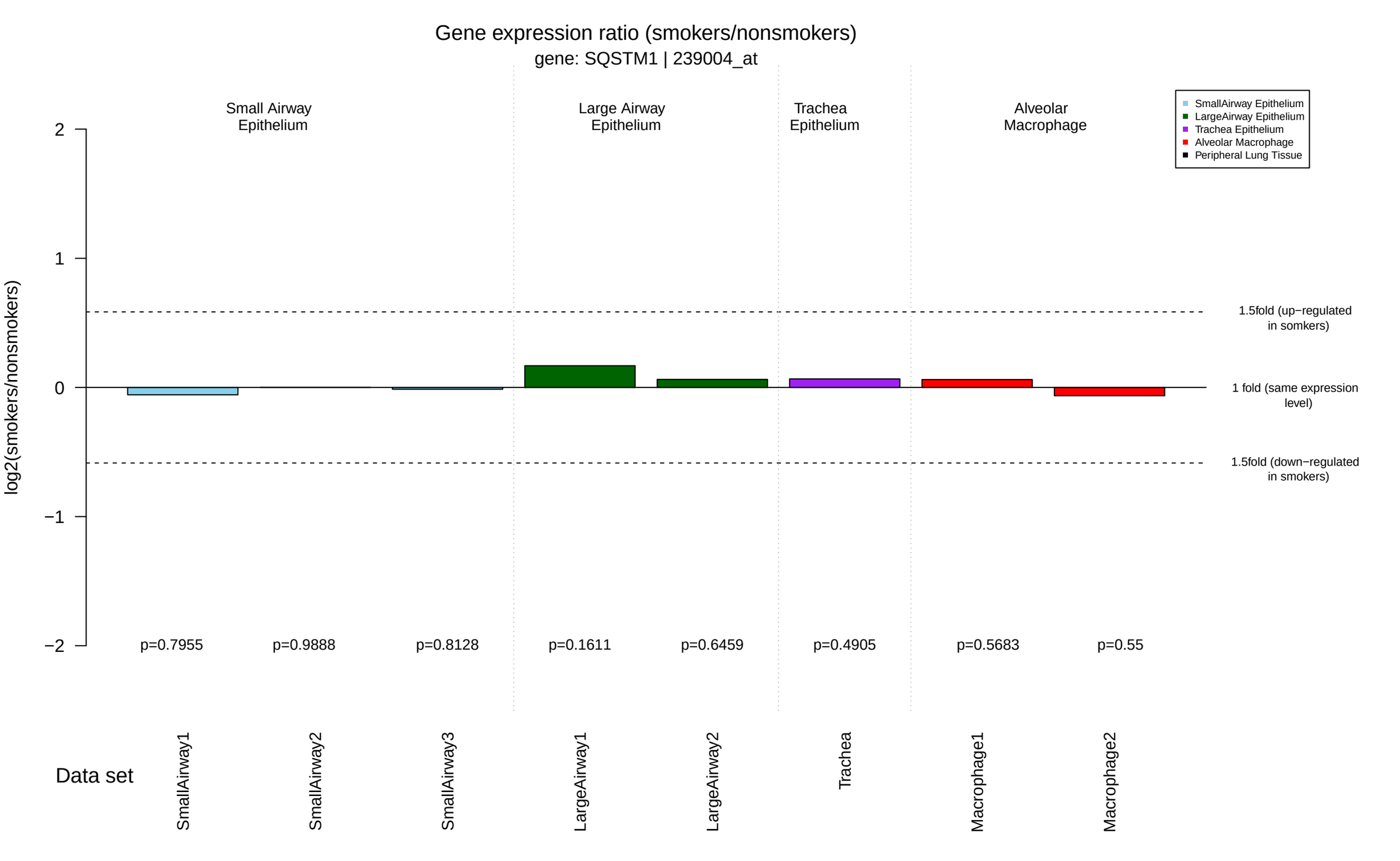
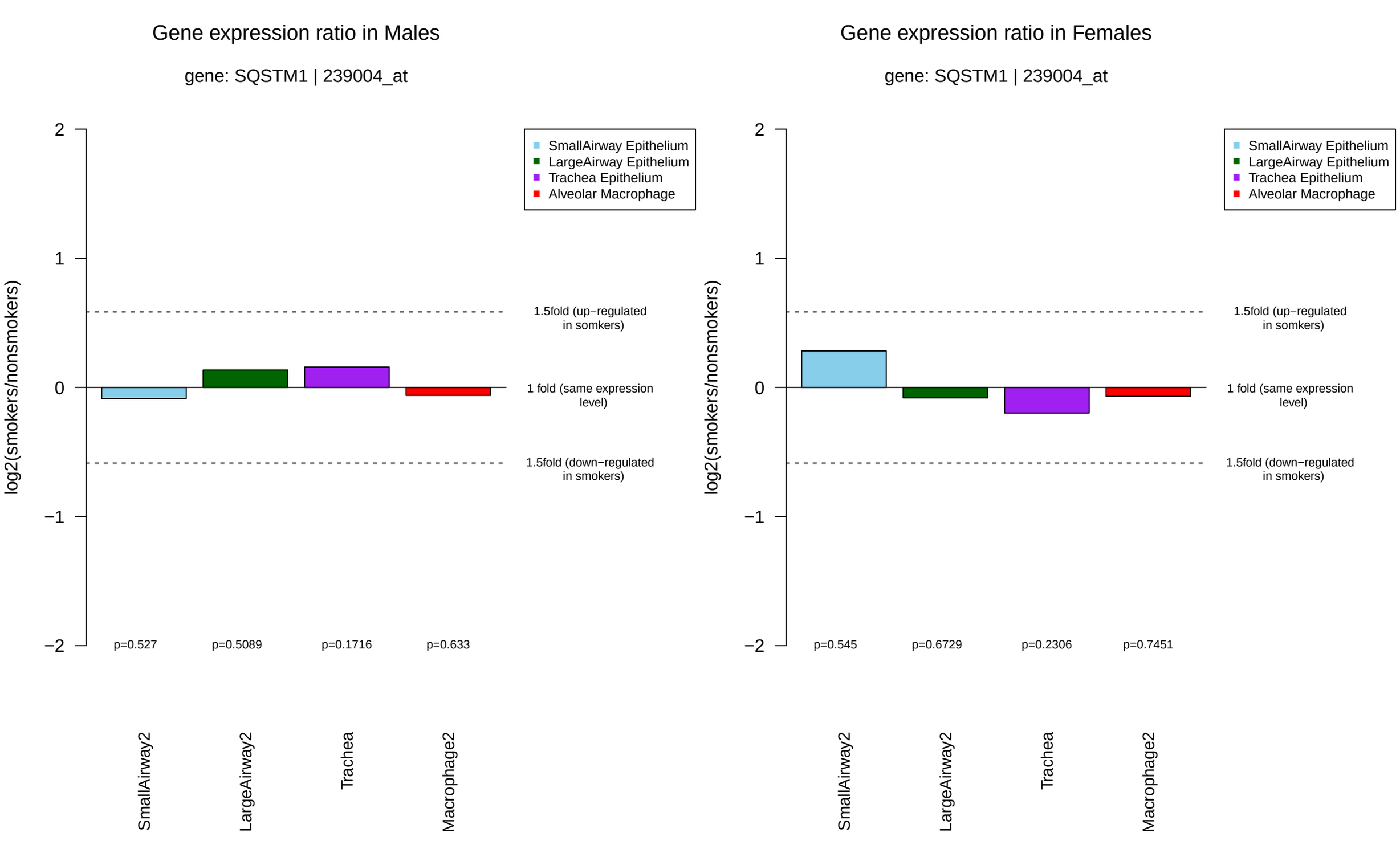
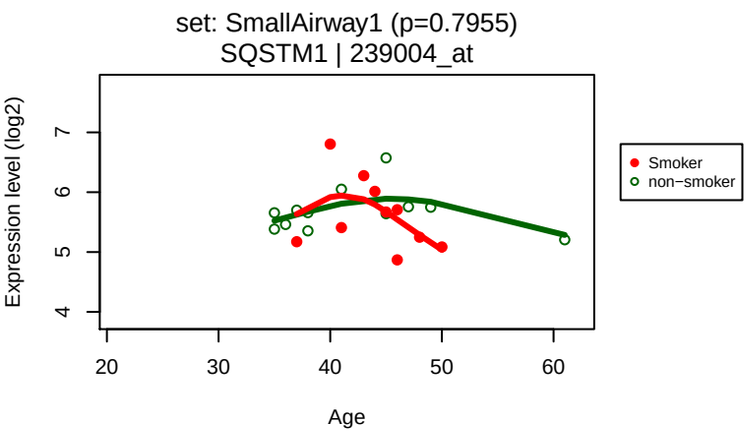
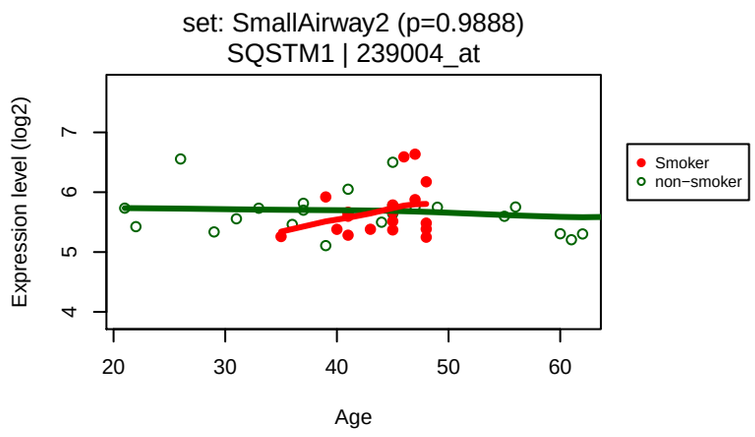
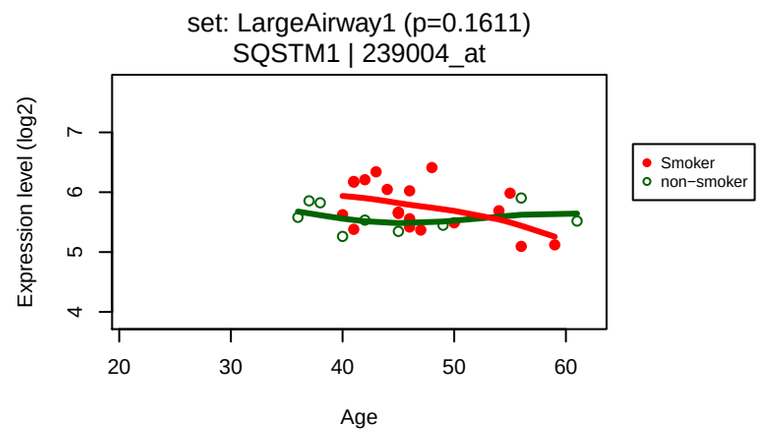

Gene expression of smokers and nonsmokers
gene: SQSTM1 | 239004_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7955
p=0.9888
p=0.8128
p=0.1611
p=0.6459
p=0.4905
p=0.5683
p=0.55
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

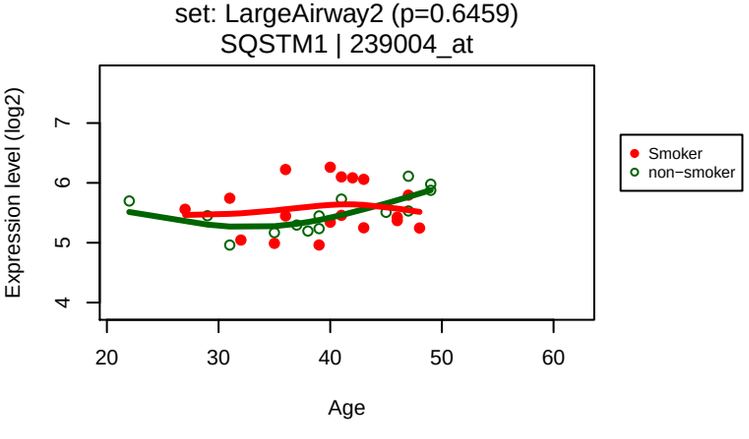

Gene expression ratio (smokers/nonsmokers)
gene: SQSTM1 | 239004_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7955
p=0.9888
p=0.8128
p=0.1611
p=0.6459
p=0.4905
p=0.5683
p=0.55
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

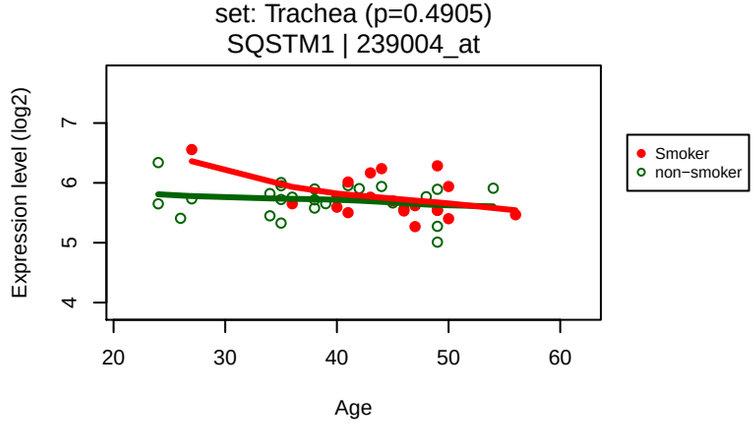

Gene expression ratio in Males
gene: SQSTM1 | 239004_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.527
p=0.5089
p=0.1716
p=0.633
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SQSTM1 | 239004_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.545
p=0.6729
p=0.2306
p=0.7451
SmallAirway2
LargeAirway2
Trachea
Macrophage2

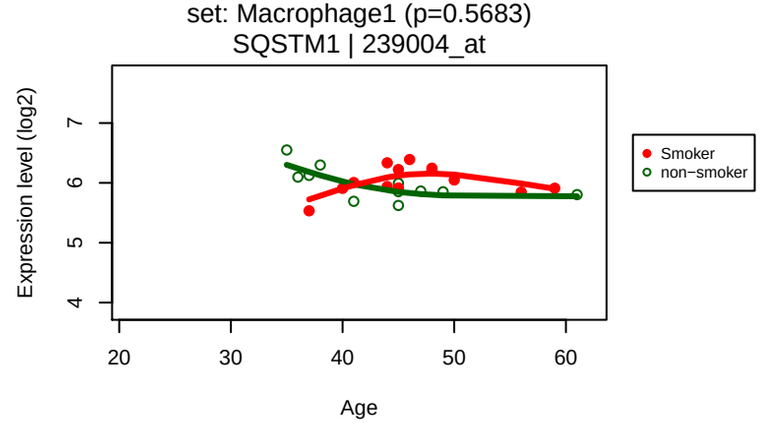

set: SmallAirway1 (p=0.7955)
SQSTM1 | 239004_at
Expression level (log2)
Age
Smoker
non−smoker

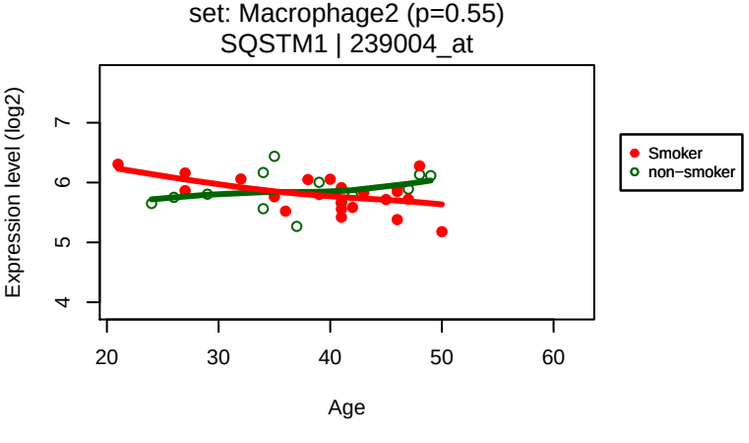

set: SmallAirway2 (p=0.9888)
SQSTM1 | 239004_at
Expression level (log2)
Age
Smoker
non−smoker

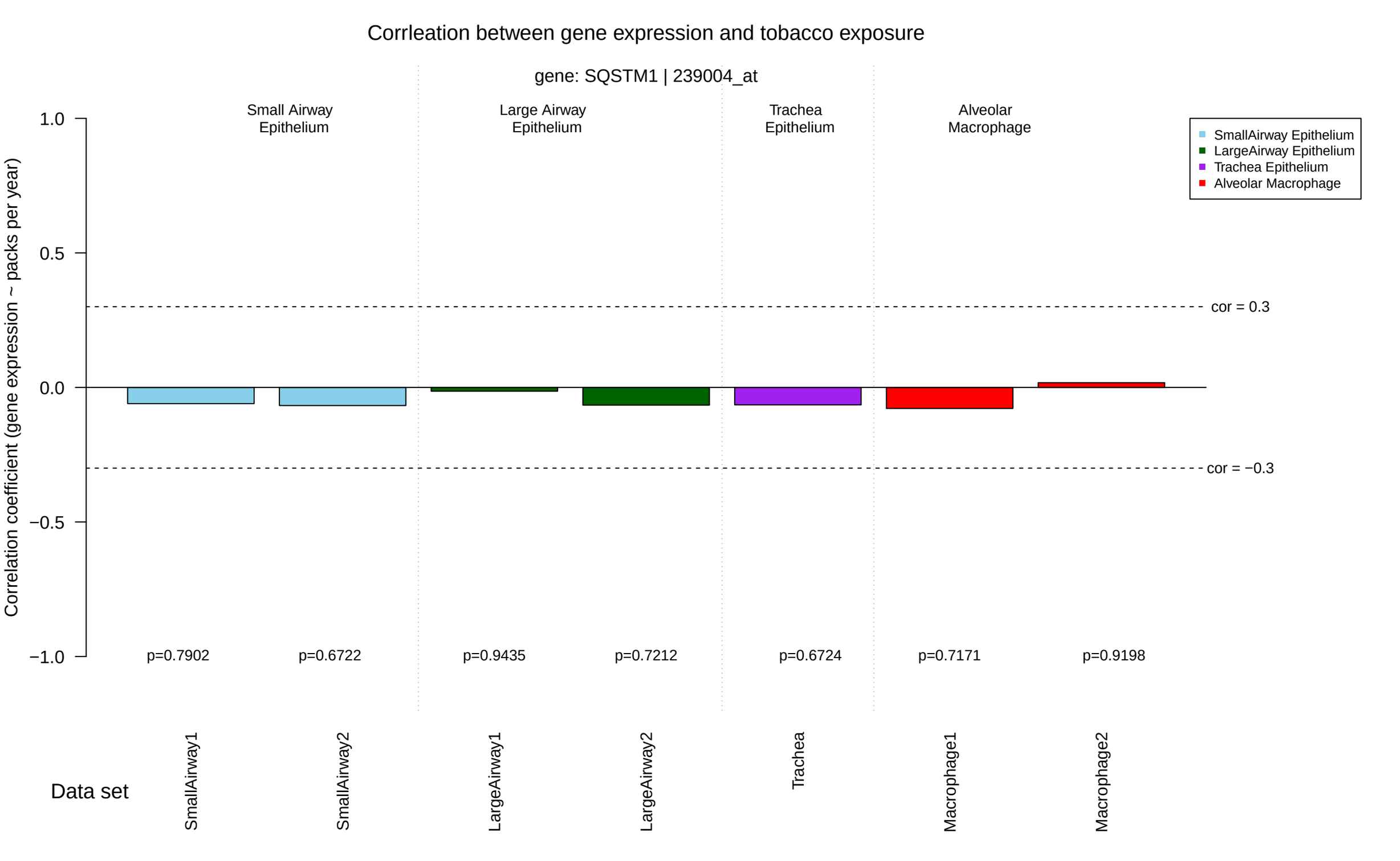

set: LargeAirway1 (p=0.1611)
SQSTM1 | 239004_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6459)
SQSTM1 | 239004_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4905)
SQSTM1 | 239004_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5683)
SQSTM1 | 239004_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.55)
SQSTM1 | 239004_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SQSTM1 | 239004_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7902
p=0.6722
p=0.9435
p=0.7212
p=0.6724
p=0.7171
p=0.9198
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2